

# Simulation and research of underground cable echo optimization based on machine learning and radar

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**Abstract**—Ground Penetrating Radar (GPR) forward modeling is crucial in interpreting real radar data and conducting full waveform inversion. To address the limitations of traditional GPR forward simulations, such as extensive computational demands and inefficiency for real-time detection, we propose a near real-time GPR forward simulation method utilizing a machine learning framework. This method is specifically designed for underground cable detection scenarios, including cable trenches, cable banks, or pipelines. The model considers cable radius and burial depth as key parameters. We employ Principal Component Analysis (PCA) to compress the echo data, deriving principal component weight coefficients. These coefficients serve as outputs for the machine learning network. Our approach features a multi-layer cyclic network architecture, coupled with a learning strategy based on Random Forest algorithms. This design aims to thoroughly examine the relationships between model parameters and principal components, enabling swift GPR simulations through machine learning. Furthermore, we integrate a Deep Neural Network with the Random Forest approach. This combination, which takes the principal component coefficients of echo data as input, establishes a scene parameter prediction model grounded in machine learning. Through inversion simulation and error analysis, we achieved a maximum depth and radius measurement error of 2 cm, with a standard deviation under 1 cm, validating the method's effectiveness.

**Keywords**—forward simulation, ground penetrating radar, underground cable, machine learning, detection

## I. INTRODUCTION

Underground cables, a critical component of urban power systems, significantly enhance the reliability and transmission capacity of city grids. These cables, integral to urban operation, planning, and management, are often likened to the "lifelines" of modern cities. As urbanization progresses, the expansion of

underground cable networks has led to frequent road excavations for installation and maintenance, posing challenges due to the aging infrastructure and inaccuracies in existing detection methods<sup>[1]</sup>.

GPR is a key tool for locating these underground cables. It works by emitting high-frequency electromagnetic waves, which are altered by the physical properties of underground structures. By analyzing these waveforms, the location and structure of subterranean objects can be identified. Enhancing GPR's real-time performance and accuracy is essential, particularly in rapid forward simulation and echo signal inversion<sup>[2]</sup>.

At present, the significance of GPR simulation research is mainly reflected in the following two aspects: forward simulation as a tool to help interpret complex real GPR data; Secondly, accurate and fast forward modeling is a key component of full waveform inversion of GPR data<sup>[3]</sup>. In addition, through radar echo simulation, the performance of GPR system can be evaluated, including detection depth, resolution, signal-to-noise ratio and other parameters. At the same time, through radar echo simulation, various parameters of GPR can be optimized. Different types of radar echo signals are generated by simulation. By applying different algorithms, simulation results can be obtained and compared with real measurement data, so as to verify the accuracy and feasibility of the algorithm<sup>[4]</sup>.

In terms of recognition accuracy of ground penetrating radar, some studies have improved the clarity of cable images by using an improved back projection algorithm, and analyzed the relationship between water pipe leakage characteristic signals and geological radar parameters<sup>[5,6]</sup>. Earlier studies proposed a new hyperbolic fitting technique to estimate the radius of buried

utilities (pipes and cables)<sup>[7]</sup>. Another study extracted the signal envelope of GPR data based on analytic signals to estimate the buried depth, diameter and dielectric constant of the underground pipeline<sup>[8,9]</sup>. In terms of radar simulation, relevant applications of machine learning methods in GPR forward simulation have been proposed. Based on Peplinks model, antenna height and surface roughness, the direct coupling response of ground penetrating radar antennas under specific soil parameters has been accurately predicted<sup>[10]</sup>. In addition, a Full Waveform Inversion method was proposed to estimate the primary characteristics of the medium, including the size and depth of rebar. This method utilized deconvolution in each iteration step and employed a mixed complex evolution algorithm to minimize the residual difference between the predicted and actual waveforms<sup>[11]</sup>. Tebchrany evaluated the efficacy of statistical clutter reduction techniques in enhancing the quality of GPR data for the detection of buried objects in soils<sup>[12]</sup>. Zhang QH presented a novel approach that integrates machine learning with fast forward simulation and buried target detection, thereby enhancing the efficiency and accuracy of GPR technology<sup>[13]</sup>. Machine learning methods, capable of delivering real-time results for complex and computationally intensive problems, have gained widespread application in GPR research in recent years. However, no existing method can precisely estimate the accuracy and deviation range of GPR. Additionally, traditional machine learning network structures and methodologies fail to accurately capture the complex interactions between the target, ground, and antenna. This limitation weakens the network's generalization capability and results in suboptimal accuracy in estimating target parameters.

In this paper, by utilizing random forest and innovative training strategies for rapid forward simulation in GPR research, a random forest based on machine learning network is proposed. Diverging from traditional machine learning network structures and methods, a composite network that integrates a neural network with a random forest, specifically for predicting parameters related to underground cables is designed. This novel approach showcases its innovation in three key aspects: Firstly, key innovations include the use of A-Scan echoes from Finite-Difference Time-Domain (FDTD) simulations as a novel data source for GPR. Secondly, Sparse Principal Component Analysis (SPCA) is introduced as the data processing method. Compared to traditional PCA, SPCA shows greater robustness in handling noisy data, capable of filtering out noise with minimal impact on principal component estimation through sparsity constraints, thereby enhancing the efficiency and accuracy of data processing. Thirdly, the proposed network is capable of sharing interrelations among principal components, a rare feature in traditional machine learning models, significantly improving the network's modeling and prediction accuracy. The inversion process of radar echo signals is completed through a machine learning architecture, validating the effectiveness of the forward modeling method proposed in this paper.

## II. SIMULATION OF UNDERGROUND CABLE ECHO BASED ON MACHINE LEARNING

This section mainly includes four parts. Firstly, the types and values of basic parameters used in the simulation process are introduced, and the sources and contents of data sets

required for forward and inversion training through machine learning are introduced. Secondly, the content of SPCA used in principal component extraction, data dimensionality reduction and noise removal is introduced. The third part introduces the design content of the network structure in the forward training of ground penetrating radar. The fourth part is mainly about the inversion process of underground cable signal through machine learning and the design method of related network.

### A. Basic parameters and data set generation

This part delves into the application of GPR for detecting underground cables within urban suburban cable channels. The focus is on a defined cuboidal detection area with dimensions of 1 m x 0.8 m x 10 m. The radar signal's interaction is detailed as illustrated in Figure 1, it first encounters a shallow soil layer with a thickness of 0.15 m, then penetrates a stone sand layer of 0.1 m thickness, and finally traverses a 0.1 m thick cover plate above the target cable. The target cable, positioned 0.1 m beneath the cover, is characterized by a diameter of 0.1 m and a length of 0.2 m. The antenna, operating at a center frequency of 1200 MHz and oriented perpendicularly to the cable's main axis, utilizes a time step of 1.925 picoseconds.:

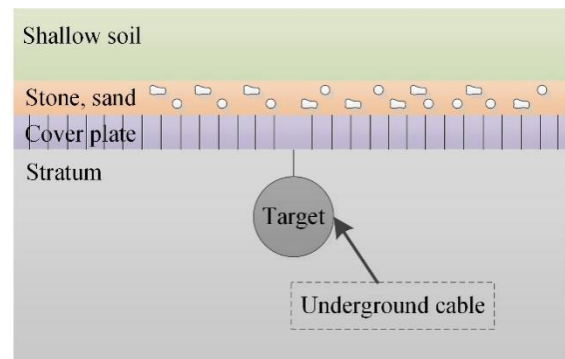


Figure 1. Underground cable laying situation

For model training, a data set is generated using numerical simulations executed in GprMax, an open-source software based on the FDTD method. This software effectively simulates the scattering echo data of targets across diverse environments and scenarios. The data set comprises 500 model samples, centering on two primary parameters: the cable's radius ( $R$ ) and its burial depth ( $D$ ). The developed learning network is capable of predicting scattering echoes in near real-time (approximately 1 second), using these parameters as inputs. This significantly expedites the GPR forward simulation process, providing robust data support for waveform verification through inversion.

### B. Sparse principal component analysis

SPCA is employed to streamline machine learning problems, specifically in the context of fast GPR forward modeling. SPCA not only accelerates the training process and reduces computational demands but also effectively filters out extraneous information from the echoes. Our training dataset comprises 1,000 scattered echoes, each containing 3,000 time steps. These echoes undergo undersampling in the time domain, adhering to the time domain sampling theorem and ensuring the efficiency of PCA. We select a sample number  $n=100$ , resulting in a dataset of 1,000 samples, each with 100 dimensions.

The covariance matrix of the training data set matrix  $T$  is set to  $\Omega$ , and the eigenvalue of  $\Omega$  is set to  $w_1 \geq w_2 \geq w_3 \geq w_4, \dots \geq w_n$ :

$$\Omega = \sum_{i=1}^m (t_i - M)(t_i - M)^T \quad (1)$$

$$M = \left(\frac{1}{m}\right) \sum_{i=1}^m t_i \quad (2)$$

Where,  $n$  is the data set dimension,  $m$  is the total number of models (i.e. the number of samples),  $t_i$  is the  $n \times 1$  dimensional column vector sample data after under sampling, and the eigenvector corresponding to  $n$  eigenvalues is  $P_n$ . After dimensionality reduction, any sample data can be composed of a series of principal components, each of which is calculated by the eigenvector matrix and the corresponding weight coefficient vector:

$$t_i = Pw_i \quad (3)$$

On this basis, in order to highlight the main parts of the principal components, display and explain the key characteristics, sparse principal component analysis method is used to solve the sparse load. The  $i$  th principal component  $F_i$  is regarded as the explained variable, and the principal component is regarded as linear regression to the  $p$  original variables, that is, the load on each variable is obtained:

$$\arg \min_{\beta} \|F_i - X\beta\|_2^2 \quad (4)$$

Where,  $X \in \mathbb{R}^{n \times p}$ ,  $\beta$  is the load vector corresponding to the  $i$  th principal component. The following Fig. 2 shows any 2 sparse principal components of the 100 feature vectors in the sample data under the scene model.

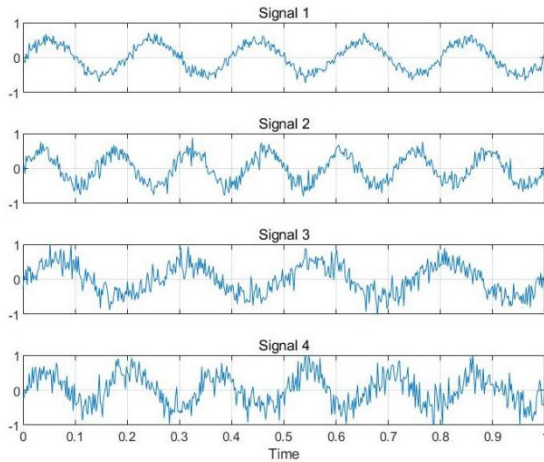


Figure 2. Example of principal component in sample data

To further expedite learning and training, we don't utilize all 100 principal components as sample data. When a specific number of linear combinations of principal components are known, the training set's echo data can be accurately fitted. Figure 3 presents the curve of the mean error between linear fitting of principal components and the echo data from FDTD numerical simulation against the number of principal components. This analysis reveals that using the first 40 out of 100 principal components achieves a highly accurate

approximation of the original echo data. Therefore, selecting these 40 principal components optimizes the balance between preserving vital information and accelerating network learning and training.

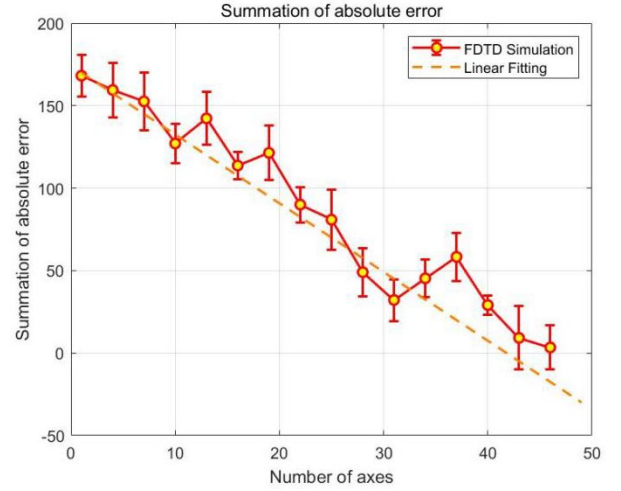


Figure 3. Error between actual and compressed A-scans

### C. Forward training network structure design

In GPR forward simulations, traditional machine learning approaches use scene parameters like cable radius and burial depth, randomly generated, as inputs. They then produce corresponding scattered echo data as outputs. These models establish the relationship between input parameters and predictive outcomes through training. However, this network design and learning methodology often fail to accurately predict GPR forward simulations in complex environments. This limitation persists despite varying network configurations, such as different numbers of hidden layers and nodes. Factors like rock and gravel interference, formation noise, delayed echo vibrations, and intricate near-field phenomena contribute to this challenge.

This part introduces a GPR forward simulation machine learning framework using a random forest approach. Random forest regression, comprising multiple decision trees, yields outputs as the average of predictions from each tree. It incorporates stochastic subspace and Bagging methods to enhance performance. The stochastic subspace method decreases tree correlation, thereby preventing overfitting. Conversely, the Bagging algorithm, by training on varied subsets of original features, circumvents local optima in regression decision trees. In this research, the random forest network structure includes 100 decision trees ( $M=100$ ). Each tree uses random variables for prediction. The overall prediction is the mean of these 100 trees' outputs, as depicted in Figure 5. A bootstrap sample  $X_k$  is produced through random sampling from the initial set, forming the basis for decision tree  $b$ . This process involves using a third of the original features to divide the sample, continuously developing the tree until it can no longer be split. The final classifier is characterized as follows:

$$a(x) = \frac{1}{N} \sum_{k=1}^M b_k(x) \quad (5)$$

Where  $a(x)$  represents the aggregate prediction for a data point with features  $x$ , computed as the average of the predictions from  $M$  decision trees within the random forest model. Each  $b_k(x)$  corresponds to the prediction made by decision tree for the input  $x$ . The summation runs from  $k=1$  to  $k=M$ , summing over all individual decision tree predictions. The division by  $N$ , which equals  $M$ , the total number of trees, normalizes this sum to produce the mean prediction.

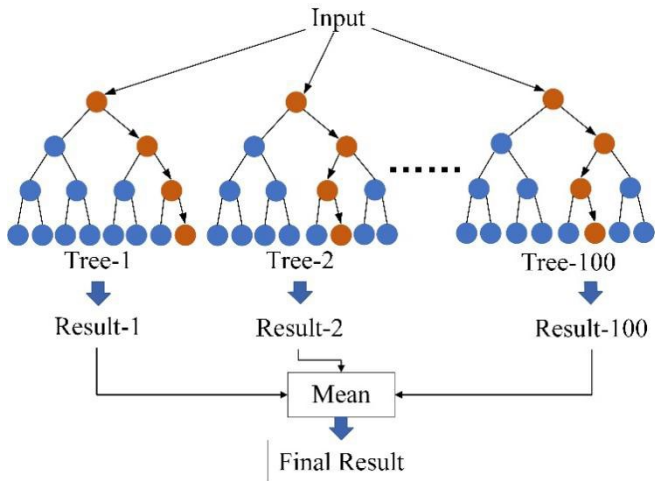


Figure 4. Basic random forest structure model

Figure 4 illustrates the random forest structure employed in this study. The model comprises multiple decision trees; for instance, it includes 100 trees (Tree-1 to Tree-100). Each decision tree receives identical input data and independently generates a result (Result-1 to Result-100). The final outcome is derived by averaging (for regression problems) or employing majority voting (for classification problems) based on the results from all individual trees. Thus, the random forest model enhances prediction accuracy, reduces model variance, mitigates overfitting, and improves performance on unseen data.

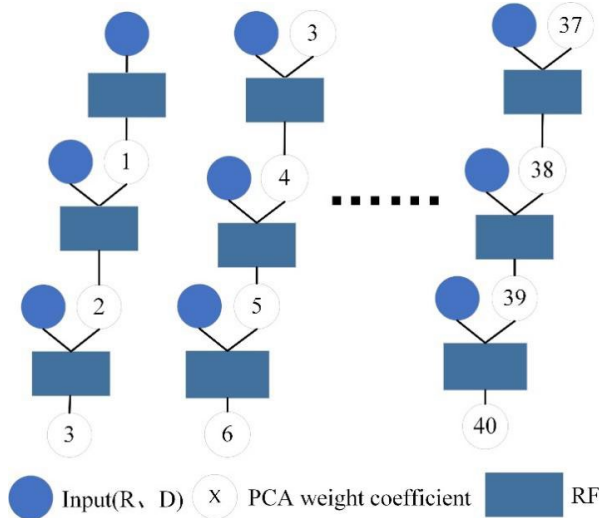


Figure 5. Fast forward modeling network architecture

The network structure and learning strategy in this paper are shown in Fig 5. The network structure consists of 40 random

forests, each with different inputs and outputs. For example, the input of the first random forest is the model parameter  $(R,D)$ , the output is the first principal component weight coefficient  $w_1$ , the input of the second random forest is the previous random forest prediction output plus the model parameter, the output is  $w_2$ , and so on, until the 40th random forest outputs  $w_{40}$ .

Each forest is individually trained to discern the causal relationships between model parameters and principal components, enhancing the network's predictive accuracy. Concurrently, these relationships constrain the search space for optimization. The culmination of this process yields the final PCA weight coefficients, and subsequent sparse PCA analysis generates sparse loads. The A-Scans echo data are then computed from these sparse loads and the eigenvectors of principal components to expedite the GPR forward simulation.

#### D. Inversion simulation network structure design

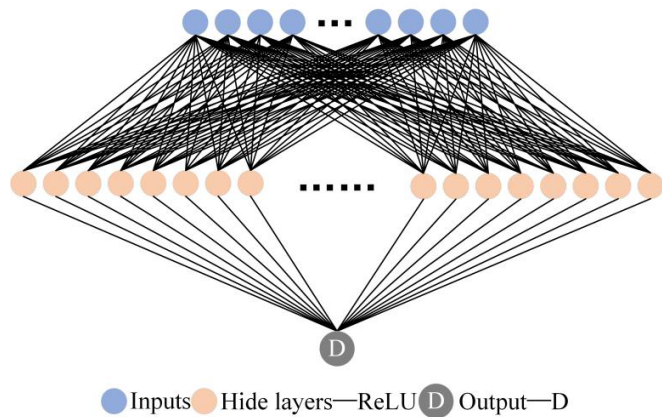


Figure 6. Depth prediction network structure diagram

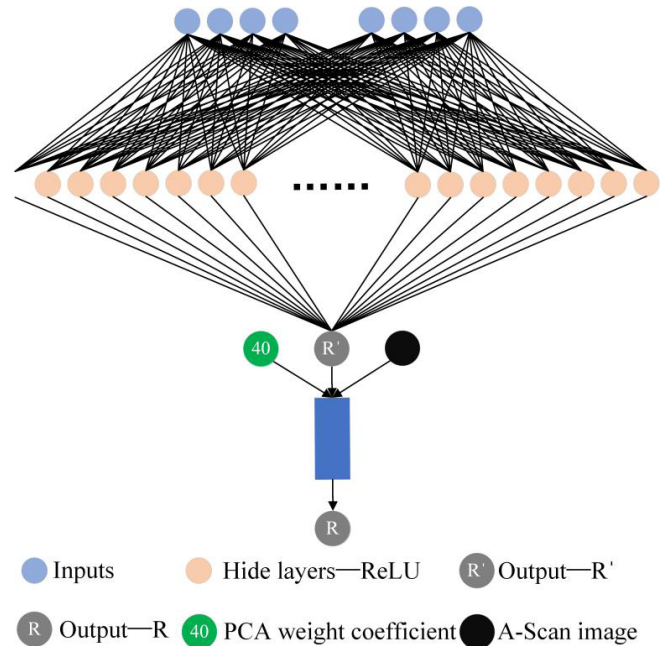


Figure 7. Radius prediction network structure diagram



Figure 6 depicts the depth prediction network architecture utilized in this study. In this architecture, the input layer is composed of numerous nodes, with each node representing an input feature. These nodes connect to several hidden layers, incorporating the nonlinear activation function ReLU, to augment the network's representational capabilities. The quantity of nodes in the hidden layers can be tailored as required to encapsulate the complexity and depth of the data. Ultimately, the output from the hidden layers is conveyed through the network and consolidated into an output node, yielding the final depth prediction. The entire network is engineered to optimize weights via training algorithms like backpropagation and gradient descent, aiming to minimize the discrepancy between predicted and actual values, thereby fulfilling the objective of depth prediction. Building on the principles outlined through the depth prediction network, this paper extends the application of machine learning methodologies to further explore the realm of underground cable detection using GPR. The neural network architecture deployed consists of a BP neural network optimized via gradient descent. As illustrated in Figures 7 and 8, the network comprises a single hidden layer with 200 nodes and utilizes a linear activation function for the output layer, while the hidden layer adopts a ReLU activation function. Input to the network is derived from 40 principal component coefficients, with the estimated buried depth (D) and cable radius (R') forming the output of two separate neural networks. To address the complexity of inferring cable radius (R), a random forest model augments the machine learning framework for enhanced predictive accuracy. Inputs to this model include principal components, A-Scans echo data, and the radius as estimated by the ANN, with the output being the radius (R). Training of the random forest results in the refined prediction of the cable radius.

### III. SIMULATION

The dataset in this paper contains 2000 model samples, 1800 are used as training samples to train and learn the above network, and 200 are used as testing samples. The PCA network prediction model was established. Fig 8 shows the comparison between the results of the conventional machine learning method based on random forest and the FDTD numerical simulation results. It can be seen that the results based on the proposed method are almost consistent with the actual GPR echo data.

Table I presents an error comparison of the prediction results from different methods across two test sets. The table includes traditional machine learning methods like Random Forest (RF) and Multilayer Perceptron (MLP) as used by Giannakis<sup>[10]</sup>, as well as Full Waveform Inversion (FWI) from Liu<sup>[11]</sup>. The proposed method demonstrates a substantial reduction in the mean square error compared to these established approaches. Notably, it improves prediction accuracy and underscores the efficacy of the proposed cyclic network structure for rapid forward simulation in Ground Penetrating Radar (GPR) applications. The results validate the 'Textual method' as a highly effective and computationally efficient alternative for numerical simulation in FWI applications.

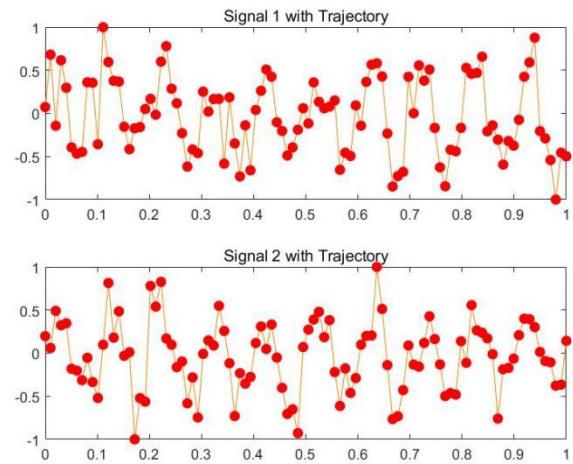


Figure 8. Comparison between the model in this paper and FDTD results

TABLE I. COMPARISON OF METHODS

|                     | Test 1      | Test 2    |
|---------------------|-------------|-----------|
| RF                  | 0.04572 m   | 0.01928 m |
| MLP <sup>[10]</sup> | 0.03540 m   | 0.01531 m |
| FWI <sup>[11]</sup> | 0.03012 m   | 0.01210 m |
| Textual method      | 0.0002489 m | 0.00085 m |

Fig 9 shows the distribution results and error histograms of the predicted and true values of the parameter network. As can be seen from Fig. 9, the prediction of depth D by the designed network is very close to the true value, the maximum error of D estimated is 1 cm, and the standard deviation is less than 0.5 cm. At the same time, it can be seen that the prediction accuracy of radius R is not high and the deviation is large. The reason for this is estimated to be due to the inherent resolution of the ground-penetrating radar antenna used.

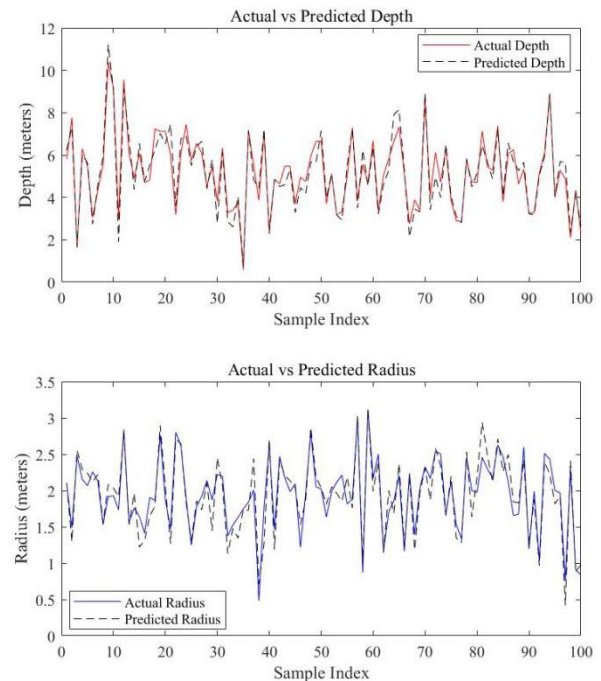


Figure 9. Comparison between the predicted values of model parameters and the true values

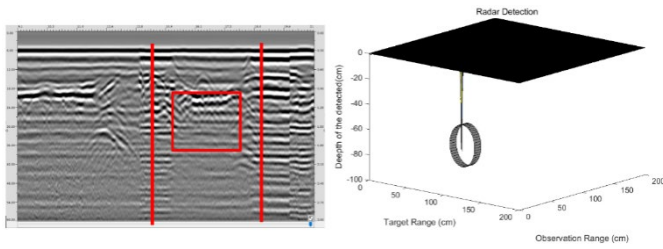


Figure 10. Comparative Imaging Analysis: Field Validation Imaging (Left) and Computational Model Imaging (Right)

Fig. 10 illustrates the effectiveness of the proposed machine learning method in accurately predicting the physical characteristics of subsurface structures. On the left, the image demonstrates the actual physical state of the underground structures as captured by in-situ imaging during field validation. This serves as an empirical benchmark for the model's predictive capabilities. On the right, the image displays the subsurface structure as reconstructed by the proposed machine learning method based on the data extracted from the field.

The side-by-side comparison of these images in Fig. 10 provides a clear visual representation of the model's performance. The empirical imaging serves as a ground truth against which the machine learning model's output is evaluated. Notably, the depth predictions are remarkably consistent with the true values, with a maximum deviation of 1 cm and a standard deviation less than 0.5 cm, suggesting high accuracy in this aspect of the model. However, the radius predictions show greater variance, highlighting an area for potential refinement of the model, likely due to the resolution limits of the radar equipment employed. This comparative analysis substantiates the simulation findings, confirming the utility of the proposed machine learning approach in real-world scenarios. By presenting both the empirical field data and the model-generated predictions, Fig. 10 validates the theoretical framework and demonstrates the practical relevance of the research, enhancing the paper's contributions to the field of subsurface detection.

The training time of the proposed method is about 4.5 minutes, and it is carried out under the computer configuration with Intel(R)Core(TM)i7-11800KF CPU and 16G memory. After the network training is completed and the model is built, the prediction results can be given in near real time (about 1 second). Considering that the network training is offline, it can be applied in real time once completed, which also reflects the effectiveness of the proposed method in terms of efficiency.

#### IV. CONCLUSION

A fast forward modeling simulator for ground penetrating radar based on machine learning is proposed. The principal components of GPR echo data simulated by 3D FDTD and reduced by PCA are used as the prediction objects of the network. The designed network structure takes random forest as the main body, including multi-layer cyclic random forest network, and adopts an innovative training and learning strategy. The intrinsic relationship between model parameters and principal components is mined to improve the modeling speed and prediction accuracy of the network. Compared with traditional machine learning methods, the designed network can

not only obtain GPR echo data in real time after offline training, but also has higher simulation accuracy. The accuracy of this method is verified by inversion. The fast forward simulation study in this paper can not only be used for machine learning-based GPR buried target detection, but also be widely used in FWI-based GPR applications as a near-real-time forward simulator.

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